

6.1GHz–7.2GHz 3W Transmitter Module (VTX)

User Manual

V1.0

2026. 05



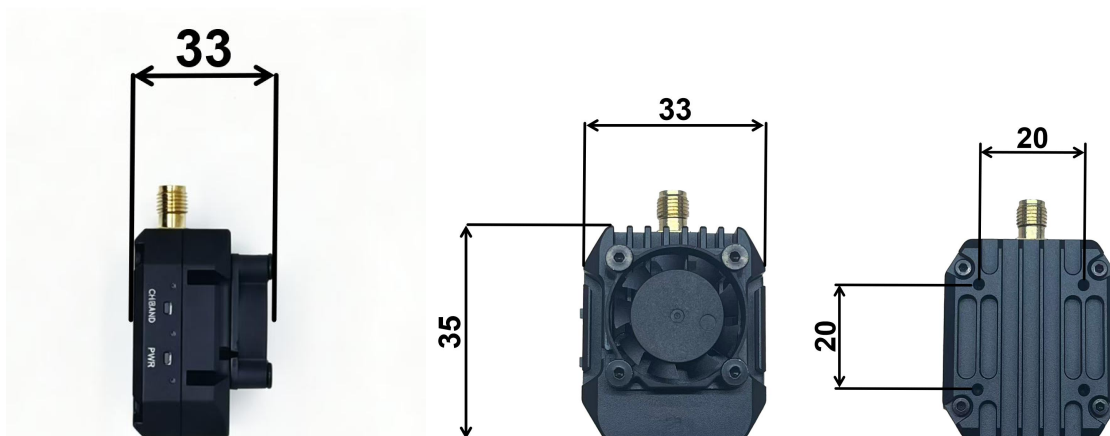
1.1 Part Name



1. SH1.0-8P cable
2. Channel Indicator (Red LED)
3. Band Indicator (Blue LED)
4. Power Indicator (Green LED)
5. Channel/ Band(CH/BAND) Button
6. Power Setting (PWR) Button
7. Antenna

1.2 Dimensions

The module dimensions and mounting dimensions are shown below (unit: mm); please reserve adequate installation clearance.



1.3 Specifications

- a. 64 channels: 6.1 GHz – 7.2 GHz (6110 MHz – 7210 MHz)
- b. Max. output power: 10 W (adjustable: 25 mW / 1 W / 2 W / 3 W)
- c. Supply voltage: DC 8 V – 36 V, 2 S–8 S Li-ion/Li-Po compatible
- d. Antenna connector: SMA female
- e. Control protocol: IRC Tramp
- f. Cooling: aluminum enclosure with heatsink and cooling fan
- g. Mounting pattern: 20 mm × 20 mm, $\varnothing$ 2.0 mm holes
- h. Weight: 56 g

2.1 Usage Instructions

2.1.1 Setting the Channel

Short-press the CH/BAND button to change the channel. Each press advances to the next channel; the corresponding red LED blinks once to indicate the selection (CH 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8, then wraps back to CH 1). See the diagram below for details.



Short-press the CH/BAND button to switch channels								
	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
RED LED	Blink once	Blinks twice	Blinks three times	Blinks four times	Blinks five times	Blinks six times	Blinks seven times	Blinks eight times

2.1.2 Setting the Band

Long-press the CH/BAND button to switch bands. Hold for 5 s: the blue LED blinks to indicate the selected band, cycling through groups A → B → E → F → R → P → H → U. Refer to the diagram below for details.



Long-press the CH/BAND button for 5s to switch bands								
	Band A	Band B	Band E	Band F	Band R	Band P	Band H	Band U
Blue LED	Blink once	Blinks twice	Blinks three times	Blinks four times	Blinks five times	Blinks six times	Blinks seven times	Blinks eight times

2.1.3 Setting Transmit Power

Press the PWR button briefly to cycle power levels: 25 mW → 1 W → 2 W → 3 W.

The green LED indicates the selected level by blinking:

- 25 mW: blinks once
- 1 W: blinks twice
- 2 W: blinks three times
- 3 W: blinks four times

See the diagram below for details.



Long-press PWR for 3s to enter Pit mode; the green LED turns solid on.

Short-press PWR to cycle power levels					
	Pit Mode	25mW	1W	2W	3W
Green LED	solid on (Long-press PWR for 3s)	Blink once	Blinks twice	Blinks Three times	Blinks four times

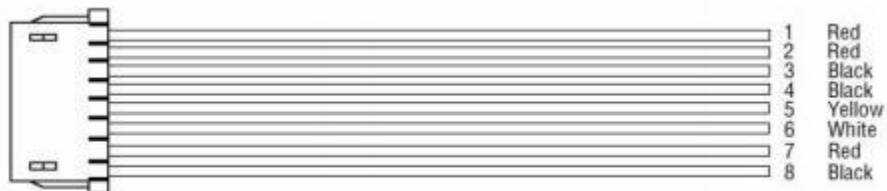
Note: This video transmitter (VTX) is equipped with thermal protection.

If the internal temperature exceeds 100 °C, output power is reduced by one step; if the temperature remains above 100 °C, it is reduced again, repeating until the minimum level (1 W) is reached. Once the temperature drops to 95 °C or below, the VTX automatically restores the originally selected power level.

2.2 Channel-to-Frequency Table

Channel-to-Frequency Table								
Band	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
Band A	6110	6130	6150	6170	6190	6210	6230	6250
Band B	6270	6290	6310	6330	6350	6370	6390	6410
Band E	6430	6450	6470	6490	6510	6530	6550	6570
Band F	6590	6610	6630	6650	6670	6690	6710	6730
Band R	6750	6770	6790	6810	6830	6850	6870	6890
Band P	6910	6930	6950	6970	6990	7010	7030	7050
Band H	7070	7090	7110	7130	7150	7170	7190	7210
Band U	6115	6265	6425	6585	6745	6905	7065	7185

2.3 SH1.0-8P Pinout Description



Pinout Description

NO	1	2	3	4	5	6	7	8
FUN	18-36V		GND		Video	IRC	5V Out	GND
Color	Red		Black		Yellow	White	Red	Black

3 Precautions

3.1 When installing the VTX, leave adequate space around the module to guarantee air-flow and proper heat dissipation. Otherwise the over-temperature protection will activate, reducing—or even shutting off—transmit power.

3.2 Before powering on, verify that the supply voltage is within the specified range and that polarity is correct to prevent damage to the unit.

3.3 Always connect an antenna to the RF output before applying power; this extends the module's service life.

3.4 Read this manual before use to ensure correct wiring and prolong the life of the module.

3.5 Do not bend the antenna cable or power leads sharply; improper handling can degrade performance or cause failure.

3.6 Understand and comply with all local laws and regulations before operating.

3.7 This product is not intended for use by children.